

HARIKRISHNAN K S

Electronics & Communication Engineering Undergraduate | Analog VLSI | Analog Layout Design | FPGA | Embedded Systems

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate interested in Analog VLSI and Layout Engineering. Gained theoretical knowledge of Analog IC Design Flow, PDKs, Physical Verification (DRC/LVS/ERC), and semiconductor design methodologies through industry training. Strong foundation in FPGA design, Verilog HDL, embedded systems, and semiconductor fundamentals with a keen interest in Analog Layout Design.

EDUCATION

B.Tech, Electronics & Communication Engineering

Expected 2027

Government Engineering College, Sreekrishnapuram, Palakkad

APJ Abdul Kalam Technological University, Kerala

CGPA : **8.14** (up to semester 5)

Higher Secondary Education - Computer Science

2021 - 2023

St. Antony's HSS Pudukkad

Percentage : **98.83%**

TECHNICAL SKILLS

Languages :

C/C++, Python, Verilog HDL

Analog VLSI :

Analog IC Design Fundamentals, **Analog Layout Concepts**, Process Design Kits (PDKs), Semiconductor Devices, **Physical Verification Concepts (DRC, LVS, ERC)**, **Parasitic Extraction (PEX)**, Corner Analysis, Analog Design Flow

FPGA & Tools :

FPGA Design, **Verilog HDL**, **RTL Design**, Vivado 2022.2, Icarus Verilog, GTKWave, Zynq-7000 (ZedBoard), **Finite State Machines (FSM)**, Synthesis, **Timing Analysis (STA)**, **Testbench Design**

Embedded Systems :

ESP32, Arduino, STM32, ESP8266, Sensor Interfacing

Hardware & Protocols :

PCB Design & Assembly, Circuit Prototyping, UART, I2C, SPI

Software & Process :

Proteus, Figma, HTML, CSS, TinkerCad, LTSpice

TECHNICAL PROJECTS

Adaptive Real-Time Traffic Light Control Using FPGA | Zedboard • FSM • Verilog • Vivado

- Developed and implemented an FPGA-based traffic control system using Verilog HDL, improving traffic flow by 25% on Zynq-7000 (ZedBoard).
- Developed FSM-based control logic to enable efficient real-time traffic signal management.
- Simulated and verified system behavior using GTKWave and Icarus Verilog.

Enviroguard – Environmental Monitoring System | STM32 • IoT • Proteus

- Designed and implemented an STM32-based environmental monitoring system, improving power efficiency by 30%.
- Integrated and calibrated sensors for temperature and pressure with real-time alerts (latency <50 ms).
- Validated circuit design and components using Proteus prior to hardware implementation.

RFID-Based Attendance System | ESP8266 • IoT

- Developed and deployed an RFID-based attendance system using ESP8266, enabling 100% contactless tracking and eliminating manual errors.
- Integrated RFID module with microcontroller for real-time data processing and monitoring.

Train Slot Booking System | GitHub for Hosting Web page • HTML • CSS

- Built and validated a Train Slot Booking System for efficient seat allocation and advance booking.
- Deployed during an industrial visit to streamline booking and reduce congestion

INTERNSHIP EXPERIENCE

Analog IC Design Trainee Intern Sophrosyne Semiconductor, Kunjibettu, Udupi	<i>June 2026</i>
• Studied the complete Analog IC Design Flow, including schematic design, layout, physical verification, and post-layout validation. • Learned PDKs, technology libraries, device libraries, simulation models, DRC, LVS, ERC, and PEX methodologies used in semiconductor design. • Explored analog building blocks such as MOSFETs, current mirrors, differential amplifiers, resistors, capacitors, and diodes. • Gained understanding of device matching, common-centroid layout, guard rings, corner analysis, Monte Carlo analysis, and reliability concepts.	
PCB Design & Assembly Intern Cyborg AI Gen Automation, Kochi, Kerala	<i>June 2025</i>
• Applied PCB design concepts including schematic capture, component selection, and layout analysis. • Assisted in circuit assembly and testing to validate hardware functionality. • Evaluated circuit designs to identify improvements in reliability and performance.	

WORKSHOPS & CERTIFICATIONS

Advanced Driver Assistance Systems & Machine Learning Workshop Indian Institute of Technology Delhi	<i>November 2023</i>
Embedded System Communication Protocols Workshop TBI - Gec Sreekrishnapuram	<i>March 2026</i>
MATLAB Workshop INVENTO 2026, Tech Fest, Gec Sreekrishnapuram	<i>June 2026</i>
Introduction to Internet of Things NPTEL	<i>November 2024</i>
Internet of Things and Embedded Systems The Arduino Platform and C Programming Interfacing with the Arduino University of California, Irvine (Coursera)	<i>June 2025</i>
Introduction to Cybersecurity Cisco Networking Academy	<i>February 2026</i>

ACHIEVEMENTS & LEADERSHIPS

- Completed multiple projects in FPGA, Embedded Systems, and Digital Design using Verilog HDL and IoT technologies.
- Active member of the Robotics Club Gec Sreekrishnapuram
- Volunteer, IEEE Young Buddies 2.0; UI/UX Designer for ECE Department.
- Design Head for ECE Department and department magazine.

LANGUAGES

Malayalam : Native **English** : Working Proficiency **Sanskrit** : Basic Proficiency

AREAS OF INTEREST

Analog VLSI • Analog Layout Design • Semiconductor Technology • Physical Verification • FPGA Design • Embedded Systems